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Experiment No. 2

Verification of KVL & KCL

Part 1: KVL

Objective

The aim of this experiment is to verify Kirchhoff's voltage law (KVL) with the help of a multi-loop circuit.

Theory

KVL stands for Kirchhoff's Voltage Law, which is a fundamental principle used in electrical engineering and physics. It states that the sum of all the voltages in a closed loop in a circuit is equal to zero (Alternatively, it can be said that around any closed circuit the algebraic sum of the voltage rises equals the algebraic sum of the voltage drops).

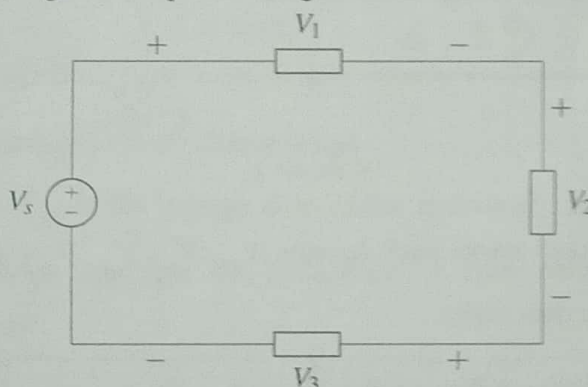


Figure 1: Illustration of KVL

To illustrate KVL, consider Fig. 1. The sign on each voltage is the polarity of the terminal encountered first as we travel around the loop. Let us start with the voltage source and go around the top, then voltages would be $-V_s + V_1 + V_2 + V_3$. Thus, KVL yields,

$$\begin{aligned}\sum \Delta V &= -V_s + V_1 + V_2 + V_3 = 0 \\ \Rightarrow V_s &= V_1 + V_2 + V_3\end{aligned}$$

Which can be interpreted as,

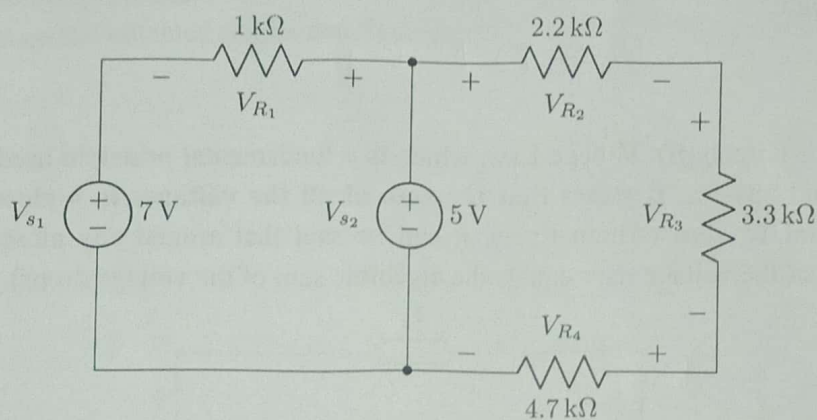
Sum of voltage rises = Sum of voltage drops

Apparatus

- Multimeter
- Resistors (1 kΩ x 2, 2.2 kΩ, 3.3 kΩ, 4.7 kΩ).
- DC power supply
- Breadboard
- Jumper wires

Procedures

- Measure the resistances of the provided resistors and fill up the data table.
- Construct the following circuit on a breadboard. Try to use minimum number of jumper wires:



Circuit 1

- Measure the voltage across each resistor (V_{R_1} , V_{R_2} , V_{R_3} , V_{R_4}) as shown in the figure above. Fill up the data tables.
- Verify KVL as $\sum \Delta V = 0$ for each loop (take the polarity of the resistors clockwise).
 For the left sided loop, $\sum \Delta V = -V_{s_1} - V_{R_1} + V_{s_2}$
 For the right sided loop, $\sum \Delta V = -V_{s_2} + V_{R_2} + V_{R_3} + V_{R_4}$
- Calculate the theoretical values of V_{R_1} , V_{R_2} , V_{R_3} , V_{R_4} and note them down in the 'Theoretical Observation' row in Table 2 & 3. For V_{R_2} , V_{R_3} , V_{R_4} use the Voltage Divider Rule. Relevant formulas are given below for your convenience:

$$V_{R_1} = V_{s_1} - V_{s_2} \quad V_{R_2} = \frac{R_2}{R_s} \times V_{s_2} \quad V_{R_3} = \frac{R_3}{R_s} \times V_{s_2}$$

$$V_{R_4} = \frac{R_4}{R_s} \times V_{s_2} \quad \text{where, } R_s = R_2 + R_3 + R_4$$

Data Tables

Signature of Lab Faculty:

S. Gais

Date:

15.6.23

**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

Table 1: Resistance Data

For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance (k Ω)
R_1	1 k Ω	0.986
R_2	2.2 k Ω	2.159
R_3	3.3 k Ω	3.250
R_4	4.7 k Ω	4.59

Table 2: Data for Loop 1 (Left sided loop)

In the following table, V_{R_1} is the voltage drop across resistor R_1 . Similar syntax applies to remaining resistors. Also, calculate the percentage of error between experimental and theoretical values of $\sum \Delta V$.

Observation	V_{s_1} (V) (from dc power supply)	V_{s_1} (V) (using multimeter)	V_{s_2} (V) (from dc power supply)	V_{s_2} (V) (using multimeter)	V_{R_1} (V)	$\sum \Delta V =$ $-V_{s_1} - V_{R_1} + V_{s_2}$ (V)
Experimental	7	7	5	5.04	-1.947	-0.053
Theoretical					1.96	0

$$\text{Absolute error} = | \text{Experimental value} - \text{Theoretical value} |$$

Here, Absolute error in $\sum \Delta V$ calculation =

0.053

Table 3: Data for Loop 2 (Right sided loop)

In the following table, V_{R_2} is the voltage drop across resistor R_2 . Similar syntax applies to remaining resistors. Also, calculate the percentage of error between experimental and theoretical values of $\sum \Delta V$.

Observation	V_{s_2} (V) (from dc power supply)	V_{s_2} (V) (using multimeter)	V_{R_2} (V)	V_{R_3} (V)	V_{R_4} (V)	$\sum \Delta V =$ $-V_{s_2} + V_{R_2} + V_{R_3} + V_{R_4}$ (V)
Experimental	5	5.04	1.085	1.634	2.315	-0.006
Theoretical			1.07	1.61	2.30	0

Here, Absolute error in $\sum \Delta V$ calculation =

0.006

Questions

- Let us take a look at **Circuit 1** again. If we remove the 5V voltage source (V_{s_2}) from the middle, the remaining circuitry contains only one big loop (often referred to as the outer loop). Let us examine if KVL holds true for the outer loop too.

- Do you think KVL will be applicable to the outer loop?

☒ Yes ☐ No

Justify your answer.

KVL can be applied on any loop so outer loop also maintains KVL.

- Use the values of V_{R_1} , V_{R_2} , V_{R_3} , V_{R_4} , V_{s_1} from Tables 2 & 3 to verify your answer from the above question.

$$\sum \Delta V = -V_{s_1} - V_{R_1} + V_{R_2} + V_{R_3} + V_{R_4} =$$

-0.019

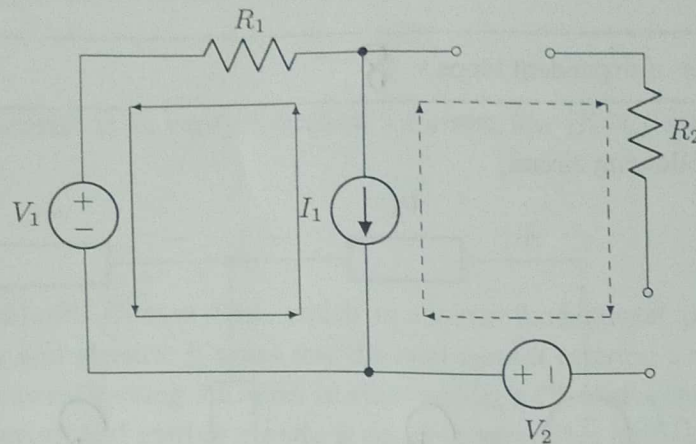
Did KVL hold true for the outer loop?

☒ Yes ☐ No

Here, absolute error in $\sum \Delta V$ calculation =

0.013

2. For the following circuit,



(a) Can we term the close path represented by the solid line made up of V_1 , R_1 , and I_1 a loop?

☒ Yes ☐ No

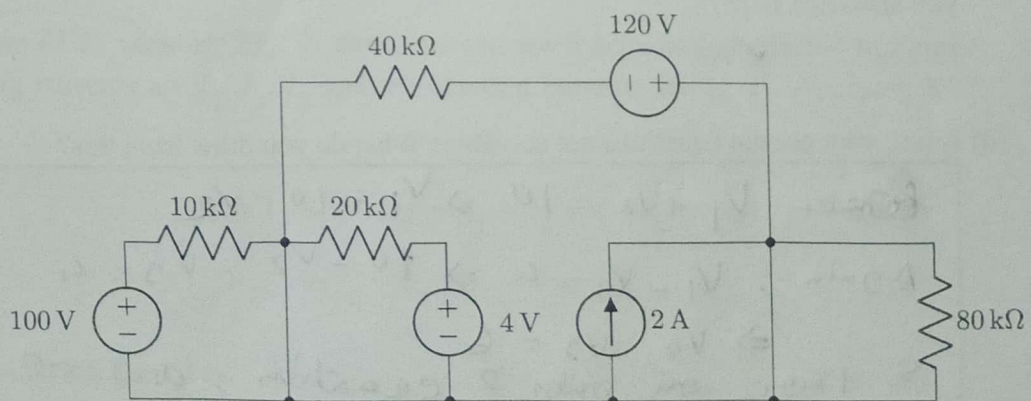
(b) Can we term the close path represented by the dashed line made up of V_2 , R_2 , I_1 , and open circuits a loop?

☐ Yes ☒ No

(c) Based on your choices in (a) and (b), how would you define a loop?

If we start from a point and return to the same point, the traversed path is considered loop.

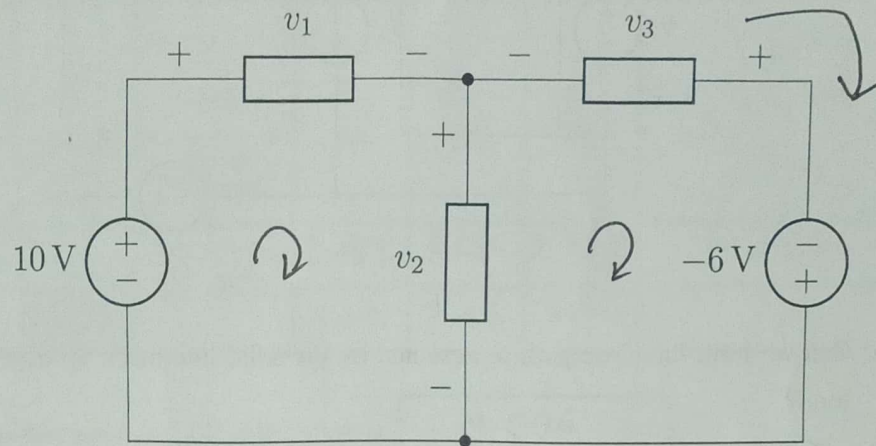
3. How many loops can you make for the following circuit? How many of them are 'Dependent' and how many are 'Independent'?



Number of independent loops = 3

Number of dependent loops = 8

4. In the following circuit,



(a) How many loops may KVL be applied along?

3

(b) List all of the equations obtained by applying KVL along the number of loops mentioned in (a).

inner loop 1 $\rightarrow -10 + v_1 + v_2 = 0$

inner loop 2 $\rightarrow -(-6) - v_2 - v_3 = 0 \Rightarrow 6 - v_2 - v_3 = 0$

outer loop $\rightarrow -10 + v_1 - v_3 - (-6) = 0 \Rightarrow -4 + v_1 - v_3 = 0$

(c) Have you been able to calculate v_1 , v_2 , and v_3 by solving the simultaneous equations you specified in (b)?

☐ Yes

☒ No

(d) If not, why are the equations not solvable? What do you draw from this?

~~inner~~ $V_1 + v_2 = 10 \Rightarrow v_1 = 10 - v_2$

Again, $v_1 - v_3 = 4 \Rightarrow 10 - v_2 - v_3 = 4$

$\Rightarrow v_2 - v_3 = 6$

So there are only 2 equations, but we

need 3 equations to get 3 unknown values.

Part 2: KCL

Objective

The aim of this experiment is to verify Kirchhoff's Current law (KCL) with the help of a simple parallel circuit.

Theory

KCL stands for Kirchhoff's Current Law, which is another fundamental principle used in electrical engineering and physics. It states that the total current entering a node in a circuit must equal the total current leaving the node. In other words, **KCL states that the algebraic sum of currents entering and exiting a node is equal to zero.** This law is also essential for analyzing circuits and predicting the behavior of electrical systems.

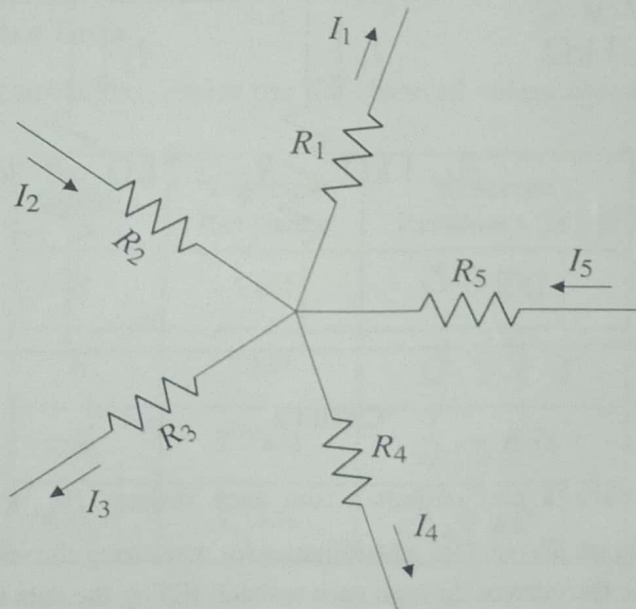


Figure 2: Illustration of KCL

To illustrate KCL, consider Fig. 2. Here, we can see 5 branches connected to 1 node. The exiting currents are I_1 , I_3 , I_4 and the entering currents are I_2 , I_5 . Applying KCL gives,

$$\begin{aligned}\sum i &= I_1 + (-I_2) + I_3 + I_4 + (-I_5) = 0 \\ \Rightarrow I_1 + I_3 + I_4 &= I_2 + I_5\end{aligned}$$

Which can be interpreted as,

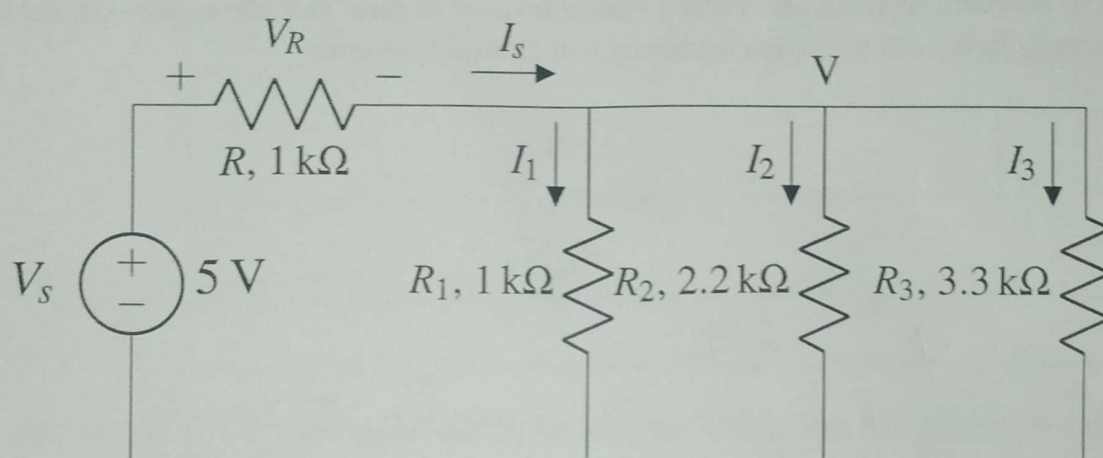
Sum of currents entering a node = Sum of currents leaving the node

Apparatus

- Multimeter
- Resistors ($1\text{ k}\Omega \times 2$, $2.2\text{ k}\Omega$, $3.3\text{ k}\Omega$, $4.7\text{ k}\Omega$).
- DC power supply
- Breadboard
- Jumper wires

Procedures

- Measure the resistances of the provided resistors and fill up the data table.
- Construct the following circuit on a breadboard. Try to use minimum number of jumper wires:



Circuit 2

- Measure the voltage and current across each resistor (V_R , V , I_s , I_1 , I_2 , & I_3) as shown in the figure above. Use a Multimeter for measuring the voltage and use Ohm's law to calculate the current through each resistor. Fill up the data tables.
- Verify KCL as $\sum i = 0$ for the node connecting R to R_1 , R_2 , & R_3 (Assume positive exiting currents).

For this node,
$$\sum i = -I_s + I_1 + I_2 + I_3$$

- Calculate the theoretical values of I , I_1 , I_2 , I_3 and note them down in the 'Theoretical Observation' row in Table 5. For I_1 , I_2 , & I_3 use the Current Divider Rule. Relevant formulas are given below for your convenience:

$$I = \frac{V_s}{R + R_p}$$

$$I_1 = \frac{(R_1)^{-1}}{(R_p)^{-1}} \times I_s \quad I_2 = \frac{(R_2)^{-1}}{(R_p)^{-1}} \times I_s$$

$$I_3 = \frac{(R_3)^{-1}}{(R_p)^{-1}} \times I_s \quad \text{where } R_p = \left((R_1)^{-1} + (R_2)^{-1} + (R_3)^{-1} \right)^{-1}$$

Data Tables

Signature of Lab Faculty:

S. Gaj

Date:

15.6.23

**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

Table 4: Resistance Data

For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance (kΩ)
R	1 kΩ	0.986
R_1	1 kΩ	0.979
R_2	2.2 kΩ	2.159
R_3	3.3 kΩ	3.250

Table 5: Data from Circuit 2

In the following table, I_1 is the current through resistor R_1 . Similar syntax applies to remaining resistors. The voltage supplied to the complete circuit is denoted by V_s and the current being supplied to the whole network is denoted as I_s . Also, calculate the percentage of error between experimental and theoretical values of $\sum i$.

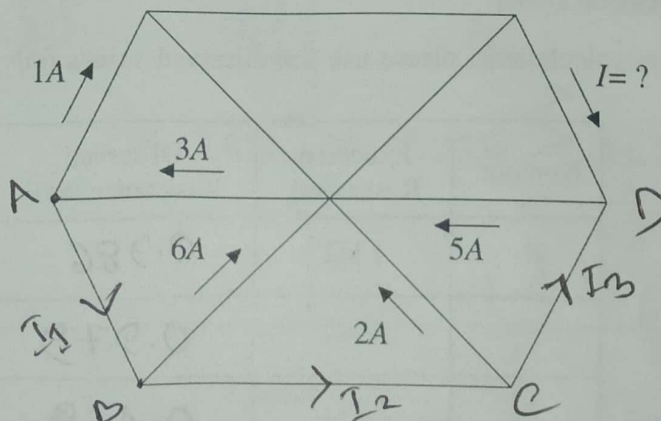
Observations	V_s (V) (from dc power supply)	V_s (V) (using multimeter)	V_R (V)	$I_s = \frac{V_R}{R}$ (mA)	V (V)	$I_1 = \frac{V}{R_1}$ (mA)	$I_2 = \frac{V}{R_2}$ (mA)	$I_3 = \frac{V}{R_3}$ (mA)	$\sum i = -I_s + I_1 + I_2 + I_3$ (mA)
Experimental	5	5.06	3.202	3.267	1.818	1.859	0.842	0.5593	0.0112
Theoretical			3.18	3.18	1.81	1.81	0.82	0.54	0

Here, Absolute error in $\sum i$ calculation =

0.0112

Questions

5. Using KCL, Determine the current I .



Node A $-3 + 1 + I_1 = 0 \Rightarrow I_1 = 2A$

Node B $-2 + I_2 + 6 = 0 \Rightarrow I_2 = -4A$

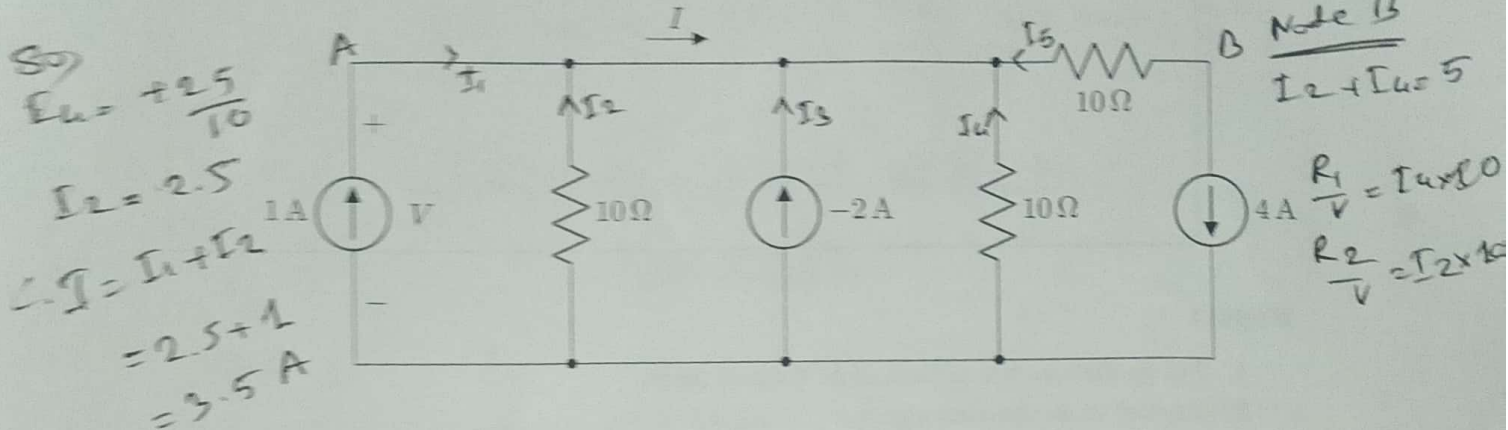
Node C $-(-4) + 2 + I_3 = 0 \Rightarrow I_3 = -6A$

Node D $-(-6) - I + 5 = 0 \Rightarrow I = 11A$

Note A $-I_1 - I_2 - I_3 - I_4 + I_5 = 0$

$\Rightarrow I_5 = 1 + I_2 - 2 + I_4$

6. For the following circuit, determine the current I using only KCL and Ohm's Law.



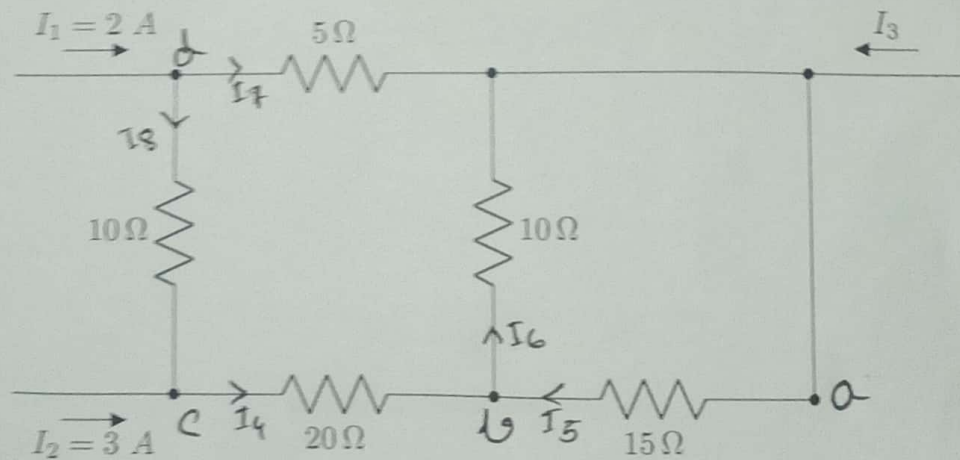
7.

(a) "KCL must always be applied at a node". The statement is-

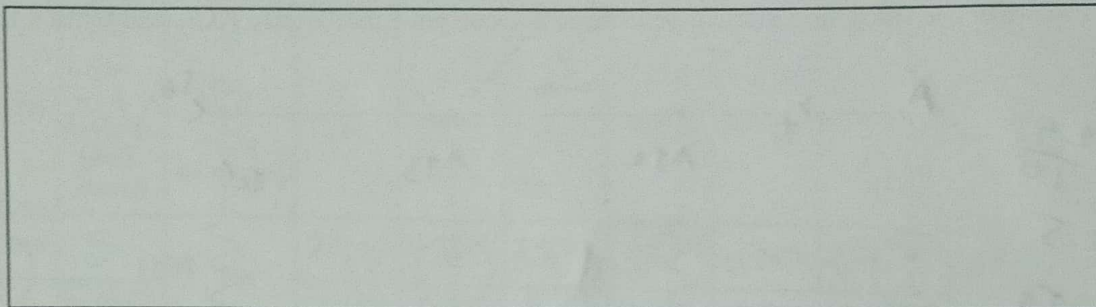
☒ True

☐ False

(b) Using KCL only, determine the value of I_3 if $I_1 = 2 \text{ A}$ and $I_2 = 3 \text{ A}$ in the circuit shown below.



<u>Note d</u> $-2 + I_7 + I_8 = 0$ $\therefore I_7 + I_8 = 2$ <u>Note a</u> $I_3 + I_7 + I_6 - I_5 = 0$ $\therefore I_3 + I_7 + I_6 = I_5$	<u>Note c</u> $-3 - I_8 + I_4 = 0$ $\therefore I_4 = 3 + I_8$ Now, $I_8 = 2 - I_7$ $\therefore I_4 = 3 + 2 - I_7$ $\therefore I_4 + I_7 = 5$	<u>Note b</u> $I_6 - I_5 - I_4 = 0$ $\therefore I_6 = I_5 + I_4$ Again, $I_3 + I_7 + I_6 = I_5$ $\Rightarrow I_3 + I_7 + I_5 + I_4 = I_5$ $\Rightarrow I_3 + I_7 + I_4 = 0$ $\Rightarrow I_3 + 5 = 0 \therefore I_3 = -5 \text{ A}$
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Report

1. Fill up the theoretical parts of all the data tables.
2. Answer to the questions.
3. Comment on the obtained results and discrepancies (if any). Start writing from here.